Title: Observing Sunspots to Measure Solar Rotation

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Introduction: Sun is a star made of He and Hydrogen gasses in Plasma state. Like other stars, the sun also rotates on its axis. Since the sun's surface isn't solid, it has different rotations depending on its regions. It is called differential rotation. Sun rotates fastest near the equator (**about 24 days**) and slowest near the poles (**around 35 days**). In 1612, Galileo first discovered that sunspots accelerate and widen as it moves closer to the center from east and slow down as it moves to the west. He thought sunspot's movement, change in size and shape is caused by its differential rotation [1]. Later in 1859, Richard Christopher Carrington observed that sun doesn't move as a rigid body. In 1863, he provided a quantitative description of differential rotation based on his observations of sunspots [2].

Because sun is made of gaseous plasma, it has different angular velocity in different latitudes [3]. Since $\mathbf{v} \propto \mathbf{r}\omega$, it moves fastest where the rotational radius is greatest (*equator*).

Like earth, sun also has its magnetic field. Because of the differential rotation, Sun's magnetic field lines get twisted. These tangled magnetic field lines come through the surface as sunspots [4]. Sunspots occur on the sun's surface in pairs for this magnetic activity. Sun's temperature is around 10,000° F. Sunspots occur in the sun's photosphere; it has a dark central region known as umbra and its surrounding is called penumbra. Umbra is about 6300° F [5]. Since Umbra is much cooler than the surroundings in photosphere, sunspots appear to be dark patches at the sun's surface - location of strong magnetic fields [6].

In this experiment we'll measure the sun's radius, observe a few sunspots over time in latitudes and compare each sunstops position changes in longitudes to measure solar rotation rate.

Methods: 1. Online Tracker, 2. Data (Sun's images) from SDO website, or from the UAlberta Observatory (white light filter telescope/Halpha, solar filter for telescope required for vision safety). 3. Scientific Calculator.

i. Since sunspots are around 6300° F (umbra). visible light range is better for sunspots observation. We used HMI Intensitygram (orange) for data collection and collected data with date and time in high resolution images and videos to save data.

ii. By using Online Tracker we first established a Heliocentric Cartesian coordinate system (x,y) keeping the center of the sun at (0,0) for the collected data (video). From the center of the Sun we measured the boundary distance and using a calibrated scale we set the radius = **696,340 km** (known value). Based on this, we measured all the data (x,y) in km.

iii. We again used the Online Tracker in tracking the sunspot's position and converted heliocentric cartesian (x,y,z) to stonyhurst heliographic cartesian(r,Θ, λ) for each sunspots.

iv. To identify the reappearance of the sunspot of our observation, we used the last heliographic longitude (before it disappeared) and subtracted 360° from it. Because longitude increases in the direction opposite the sunspot's motion (westward). Since our coordinate system is set up such that x is positive to the right when facing the Sun and the sunspots are moving rightward (west to east), we subtracted 360° from its longitude.

v. (21 - 23 March) We calculated the angular velocity of the sunspots from change in position and time. Then determined the pattern in rotation.

vi. (24 - 30 March) We will also consider the size of the sunspots. Since sunspots aren't always round in shape. We will measure the sunspots area using ImageJ. In ImageJ we will open sunspots' images, use process > binary > make binary to get binary representation of the sunspots. Then select analyze > analyze particles. From the pop-up menu we will select show; outlines

(detected region), display results and click ok. We will get sunspots area in pixels. Now we will observe graph of angular velocity vs $sin^2(latitude)$ graph [7, 8]. Now, we will analyze the data to understand sunspots' contribution in solar rotation.

vii. **(1 - 3 April)** Prepare and submit final report.

viii. **(4 - 11 April)** Prepare slides for the presentation and presenting to groups.

Heliocentric Cartesian to Stonyhurst Heliographic Cartesian [9]:

$x = r\,sin\Theta\,cos\lambda;$ $\quad z = r\,cos\Theta\,;$ $\quad r = \sqrt{x^2 + y^2 + z^2}\,;$

$y = r\,sin\Theta\,sin\lambda;$ $\quad \Theta = cos^{-1}(\frac{z}{r})\,;$ $\quad \lambda = tan^{-1}(\frac{y}{x})$

x: The pixel measurement along the x-axis (horizontal) from the center of the Sun.
y: The pixel measurement along the y-axis (vertical) from the center of the Sun.
z: The distance from the Sun's center to the sunspot. Since the sunspots distance from the center = sun's radius (it's on the surface of the Sun), we will assume **r = 696,340 km.**
r: is the radial distance from the Sun's center to the sunspot.
Θ , λ: Stonyhurst heliographic latitude and longitude.

Calculation: **Rotation period for the sunspots:**
Time of image 1 (hh-mm-ss) = Time of image 1 (hh)
Time of image 2 (hh-mm-ss) = Time of image 1 (hh)
Time of image 1 (hh) - Time of image 2 (hh) = ΔTime
ΔTime/24 = x days
(Day of image 1 - day of image 2) = Days between images
Days between images + x days = time difference between image 1 and 2 = ΔTiime interval

Average Latitudes = Latitude of image 1 - Latitude of image 2
|ΔLongitude| = |Longitude of image 1 - Longitude of image 2|
[Same process will be repeated for all the pairs]

Pair (1,2): at average latitude, to complete ΔLongitude (degrees) time taken = ΔTiime interval
to complete one rotation (360 degrees) time taken = $\frac{\Delta Tiime\ interval}{|\Delta Longitude|}360$ = y days [Period (1,2)]
[Same process will be repeated for all the sunspots]

Sunspots' data:

Sunspot 1 (close to the equator)	Date and time (dd-mm-yy, hh:mm:ss)	x (km)	y (km)	r (km)	Sunspot's Latitude, Θ (degrees)	Sunspot's Longitude, λ (degrees)
1	14-01-23, 04:30:00	-6.33E+05	-1.47E+05	952463.855	43.02	193.07
2	15-01-23, 19:15:00	-5.01E+05	-1.35E+05	868398.178	36.69	195.08
3	16-01-23, 05:15:00	-4.56E+05	-1.28E+05	842145.709	34.22	195.68
4	17-0123, 03:45:00	-3.36E+05	-1.23E+05	782888.495	27.20	200.11
5	18-01-23, 23:30:00	-6.69E+04	-1.17E+05	709263.002	10.95	240.24

6	19-01-23, 22:00:00	7.90E+04	-1.17E+05	710506.436	11.46	304.03
7	**20-01-23, 10:15:00**	**1.56E+05**	**-1.17E+05**	**723128.201**	**15.64**	**323.13**
8	21-01-23, 08:30:00	2.92E+05	-1.22E+05	764877.373	24.44	337.32
9	22-01-23, 20:30:00	4.76E+05	-1.36E+05	854377.783	35.41	344.05
10	23-01-23, 10:45:00	5.36E+05	-1.42E+05	890140.099	38.53	345.16

Sample Table Calculations:

For sunspot 1 at that position 7,

r = $\sqrt{(1.56E+05)^2 + (-1.17E+05)^2 + (696,340)^2}$ = 723128.201

$\Theta = cos^{-1}(\frac{z}{r}) = cos^{-1}(\frac{696,340}{723128.201}) = 15.64°$

λ = $tan^{-1}(\frac{-1.17E+05}{1.56E+05})$ = 345.16°; [repeating same process to calculate r, Θ and λ for all the sunspots]

image Pair	ΔTime interval (days)	Average Latitude (degrees)	ΔLongitude (degrees)	Period (days)
(1,2)	0.3850			
(3,4)	1.0625			
(5,6)	1.0625			
(7,8)	**1.0729**	**-8.8**	**14.19**	**27.2195**
(9,10)	1.4063			

Sample Calculation: Rotation period for the sunspots:

Time of image 7 (10:15:00) = Time of image 7 (10.25:00:00)

Time of image 8 (08:30:00) = Time of image 8 (8.50:00:00)

Time of image 7 (10.25) - Time of image 8 (8.50) = 1.75 hrs

1.75/24 = 0.0729 days

(Day of image 7 - day of image 8) = 20 - 21; Days between images = 1

ΔTiime interval = 1 + 0.0729 days = 1.0729 days

Average Latitudes = Latitude of image 7 - Latitude of image 8 = 15.64 - 24.44 = -8.8

|ΔLongitude| = |Longitude of image 7 - Longitude of image 8| = |**323.13 -** 337.32| = |-14.19| = 14.19

[Same process will be repeated for all the pairs]

Pair (7,8): at average latitude, to complete ΔLongitude (degrees) time taken = **1.0729** days

to complete one rotation (360 degrees) time taken = $\frac{1.0729}{14.19}360$ = 27.2195 days [Period (1,2)]

[Same process will be repeated for all the sunspots]

Angular Velocity Calculation:

ω = $\frac{v}{r}$; ω : angular velocity; v: linear velocity of the sunspot ,

r: is the radial distance from the Sun's center to the sunspot = 696,340 km

Sunspot no.	r (km)	λ_{rad} (radians)			ΔTime interval (days)	Angular Velocity $\omega = \frac{v}{r}$ (rad/day)
1						

Error and Bias:

1. We used Online Tracker and collected data in high resolution (images/video) to minimize the error in tracking the sunspot's position. Also, we kept backup images (data) to work.
2. We repeated the process a few times for each sunspot and took the average value for calculations to reduce error and biases.
3. For the size consideration part, we need to find at least two sunspots with greater difference in size. For that we might need to collect data outside of the boundary (12 - 15 Feb, 2024) of selected dates. If any sunspot is outside of the date boundary we will mention it in the report.
4. Sunspots could shift in latitudes over time. For that we will take all the latitudes data and consider the average latitude for the experiment.
5. We will do error propagation in each step of data analysis to minimize the error.

Result: Expected results are -

1. Sun rotates fastest near the equator (about 24 days) and slowest near the poles (around 35 days).
2. The sun has different angular velocity in different latitudes and it moves fastest where the rotational radius is greatest (*equator*).
3. Larger (area) sunspots rotate slower than sunspot's with smaller area.

Citations:

[1] NASA. "Solar Rotation Varies by Latitude." *NASA*, 23 Jan, 2013. Accessed 20 Feb, 2024, https://www.nasa.gov/image-article/solar-rotation-varies-by-latitude/

[2] Obridko, V.N. and Badalyan, O.G. (2019) Solar Corona as Indicator of Differential Rotation of Subphotospheric Layers. *Cosmic Research*, 57, 407-412.

[3] "The Rotation of the Sun." *Bristol University*. Accessed 20 Feb. 2024, https://www.star.bris.ac.uk/bjm/solar/solarrot.html#:~:text=The%20rotation%20of%20the%20Sun,and%20decreases%20as%20latitude%20increases.

[4] National Earth Science Teachers Association. "Sunspot Magnetism." *Windows to the Universe*, https://www.windows2universe.org/sun/atmosphere/sunspot_magnetism.html#google_vignette, Accessed 21 Feb, 2024.

[5] National Weather Service. "Sunspots." *National Oceanic and Atmospheric Administration*, https://www.weather.gov/fsd/sunspots. Accessed 19 Feb, 2024.

[6] Weisstein, Eric W. "Sunspot." *Eric Weisstein's World of Astronomy, Wolfram Research*, 2007, http://scienceworld.wolfram.com/astronomy/Sunspot.html, Accessed 19 Feb, 2024.

[7] Stenflo, J. O., "Differential rotation of the sun's magnetic field pattern" *Astronomy and Astrophysics*, vol. 210, no. 1-2, Feb. 1989, p. 403-409, https://adsabs.harvard.edu/full/1989A%26A...210..403S

[8] Jha, Bibhuti Kumar et al. "Measurements of Solar Differential Rotation Using the Century Long Kodaikanal Sunspot Data." Solar Physics. ResearchGate. Accessed 20 Feb. 2024, https://www.researchgate.net/figure/Rotation-profiles-of-sunspots-with-area-200-Hem-blue-dashed-line-and-with-area-400_fig2_348294445

[9] Paul. "Spherical Coordinates." *Paul's Online Notes*, https://tutorial.math.lamar.edu/classes/calciii/SphericalCoords.aspx.

<u>Sample Table Calculations:</u>

Average y = -1.28E+05

$sin\theta = \frac{average\ y}{R} = \frac{-1.28E+05}{696{,}340}$

= -1.84E-01, hence θ

$= sin^{-1}$ (-1.84E-01) = -10.60°

x(t) = $Rcos\theta sin(\omega t + \phi)$;

-6.33E+05 = $696340cos(-10.60°)sin(\omega(t) + \phi)$;

I plotted the data in python and adjusted the value of ω and ϕ to check for the best fit line.

ω = 0.25 ± 0.1 and ϕ = 2.5 ± 0.1 is the value for the best fit line.

<u>Fig.1</u>: I plotted the x value along the y axis and time (t) along the x axis for the equation x(t) = $Rcos\theta sin(\omega t + \phi)$; Here **x**: The distance measurement along the x-axis (horizontal) from the center of the Sun in km. R: Sun's radius 696,340 km, θ = -10.60° (sample calculation), checked for different values of ω : angular velocity and ϕ = phase.